

SF1811 Optimization

Lecture 9: Quadratic optimization with equality constraints

Chapter 10 and selected parts of Chapter 25

Anders Forsgren

Quadratic optimization with equality constraints

Main question:

How do equality constraints change the curvature and optimality conditions of a quadratic problem?

Today:

- Feasible points and feasible directions.
- A nullspace representation of the feasible set.
- The reduced Hessian.
- The nullspace method.
- The Lagrange method.
- Comparison of the two methods.

Quadratic optimization with equality constraints

Consider

$$(P) \qquad \begin{aligned} &\text{minimize} && f(x) = \tfrac{1}{2} x^{\mathsf{T}} H x + c^{\mathsf{T}} x + c_0 \\ &\text{subject to} && Ax = b, \end{aligned}$$

where

$$H = H^{\mathsf{T}} \in \mathbb{R}^{n \times n}, \qquad A \in \mathbb{R}^{m \times n}.$$

We assume that

- A has full row rank;
- the system $Ax = b$ has at least one solution;
- $m < n$, so that there are nonzero feasible directions.

The feasible set is the affine set

$$F = \{x \in \mathbb{R}^n : Ax = b\}.$$

The objective need only have nonnegative curvature along feasible directions.

Feasible points and feasible directions

Let $\overline{x}$ be a known feasible point: $A\overline{x} = b$.

A point x is feasible if and only if $Ax = b = A\overline{x}$, which is equivalent to

$$A(x - \overline{x}) = 0.$$

Thus

$$x - \overline{x} \in \mathcal{N}(A).$$

Similarly, a direction d preserves feasibility if and only if

$$A(\overline{x} + td) = b$$

for every t, which is equivalent to

$$Ad = 0.$$

Hence

$$\boxed{d \text{ is a feasible direction} \iff d \in \mathcal{N}(A).}$$

A nullspace representation of the feasible set

Let the columns of

$$Z \in \mathbb{R}^{n \times (n-m)}$$

form a basis for $\mathcal{N}(A)$.

Then

$$AZ = 0,$$

and every vector in $\mathcal{N}(A)$ has a unique representation Zv.

Therefore,

$$\boxed{x \in F \quad \Longleftrightarrow \quad x = \overline{x} + Zv \text{ for a unique } v \in \mathbb{R}^{n-m}.}$$

Constructing a nullspace matrix

After ordering the variables, partition

$$A = (A_B \; A_N),$$

where $A_B \in \mathbb{R}^{m \times m}$ is nonsingular.

Partition a vector $d \in \mathcal{N}(A)$ accordingly: $\quad d = \begin{pmatrix} d_B \\ d_N \end{pmatrix}$.

Then

$$Ad = 0 \quad \Longleftrightarrow \quad A_B d_B + A_N d_N = 0,$$

so that

$$d_B = -A_B^{-1} A_N d_N.$$

Setting $v = d_N$ gives

$$d = \begin{pmatrix} -A_B^{-1} A_N \\ I \end{pmatrix} v = Zv, \quad \text{where} \quad Z = \begin{pmatrix} -A_B^{-1} A_N \\ I \end{pmatrix}.$$

Why does this matrix span the nullspace?

For

$$Z = \begin{pmatrix} -A_B^{-1} A_N \\ I \end{pmatrix},$$

we have

$$AZ = (A_B \; A_N) \begin{pmatrix} -A_B^{-1} A_N \\ I \end{pmatrix} = -A_N + A_N = 0.$$

Thus every column of Z belongs to $\mathcal{N}(A)$.

The identity block shows that the columns of Z are linearly independent. Since

$$\dim \mathcal{N}(A) = n - m,$$

the columns of Z form a basis for $\mathcal{N}(A)$.

This is the same nullspace representation that appeared implicitly in the simplex method.

A small nullspace example

Let $A = \begin{pmatrix} 1 & 1 & 1 \end{pmatrix}$. Choose

$$A_B = (1), \qquad A_N = \begin{pmatrix} 1 & 1 \end{pmatrix}.$$

Then

$$Z = \begin{pmatrix} -1 & -1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Indeed,

$$AZ = \begin{pmatrix} 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} -1 & -1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \end{pmatrix}.$$

Thus every solution of $x_1 + x_2 + x_3 = b$ can be written as $x = \overline{x} + Zv$.

The reduced quadratic problem

Substitute

$$x = \overline{x} + Zv$$

into the exact quadratic expansion:

$$\begin{aligned} f(\overline{x} + Zv) &= f(\overline{x}) + (H\overline{x} + c)^{\mathsf{T}} Zv + \frac{1}{2}(Zv)^{\mathsf{T}} H(Zv) \\ &= f(\overline{x}) + \big(Z^{\mathsf{T}}(H\overline{x} + c)\big)^{\mathsf{T}} v + \frac{1}{2} v^{\mathsf{T}}(Z^{\mathsf{T}} HZ)v. \end{aligned}$$

Hence the equality-constrained problem in x is equivalent to the unconstrained problem

$$\underset{v \in \mathbb{R}^{n-m}}{\text{minimize}} \quad f(\overline{x}) + \big(Z^{\mathsf{T}}(H\overline{x} + c)\big)^{\mathsf{T}} v + \frac{1}{2} v^{\mathsf{T}}(Z^{\mathsf{T}} HZ)v.$$

Equality constraints reduce the problem to an unconstrained quadratic problem in $n - m$ variables.

The reduced Hessian

The Hessian of the reduced problem is

$$\boxed{Z^{\mathsf{T}} H Z.}$$

Therefore,

$$f \text{ is convex on } F \quad \Longleftrightarrow \quad Z^{\mathsf{T}} H Z \succeq 0,$$

and

$$f \text{ is strictly convex on } F \quad \Longleftrightarrow \quad Z^{\mathsf{T}} H Z \succ 0.$$

The condition $H \succeq 0$ is sufficient, but not necessary, for convexity on F.
Indeed, only directions of the form

$$d = Zv \in \mathcal{N}(A)$$

are feasible, and along such directions

$$d^{\mathsf{T}} H d = v^{\mathsf{T}} Z^{\mathsf{T}} H Z v.$$

Only the curvature along feasible directions matters.

Convex on the feasible set, but not in the full space

Consider

$$H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \qquad A = \begin{pmatrix} 0 & 1 \end{pmatrix}, \qquad b = 0.$$

The matrix H is indefinite, so the quadratic function is not convex on $\mathbb{R}^2$.

However, the constraint is

$$x_2 = 0,$$

so the feasible directions are multiples of

$$Z = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

The reduced Hessian is

$$Z^{\mathsf{T}} H Z = \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 1 > 0.$$

Thus the problem is strictly convex on the feasible set.

The nullspace method

Assume that

$$Z^T H Z \succeq 0.$$

The reduced problem is minimized when its gradient is zero:

$$(Z^T H Z)\hat{v} = -Z^T(H\overline{x} + c).$$

The corresponding solution to the original problem is

$$\boxed{\hat{x} = \overline{x} + Z\hat{v}.}$$

If

$$Z^T H Z \succ 0,$$

then the reduced system has a unique solution and $\hat{x}$ is the unique minimizer.

Existence and uniqueness in the nullspace method

The reduced problem has exactly the same alternatives as an unconstrained quadratic problem:

- If $Z^T H Z \not\succeq 0$, the objective is unbounded below along a feasible direction.
- If $Z^T H Z \succeq 0$ but the reduced optimality equation has no solution, the objective is unbounded below.
- If the reduced optimality equation has a solution, one or more minimizers exist.
- If $Z^T H Z \succ 0$, the minimizer is unique.

The matrix $Z^T H Z$ replaces H in the constrained problem.

Feasible descent directions

Assume that

$$Z^{\mathsf{T}} H Z \succeq 0,$$

so that the problem is convex on the feasible set.

A vector d is a feasible direction if and only if

$$d \in \mathcal{N}(A).$$

For a feasible direction d, the exact expansion is

$$f(x + td) - f(x) = t(Hx + c)^{\mathsf{T}} d + \frac{1}{2} t^2 d^{\mathsf{T}} H d.$$

Since $d^{\mathsf{T}} H d \geq 0$ for every feasible direction,

$$d \text{ is a feasible descent direction at } x \quad \iff \quad (Hx + c)^{\mathsf{T}} d < 0.$$

Optimality means orthogonality

At an optimal solution $\hat{x}$, there is no feasible descent direction.

For every $d \in \mathcal{N}(A)$, both d and $-d$ are feasible directions. Therefore,

$$(H\hat{x} + c)^{\mathsf{T}} d \geq 0$$

and

$$(H\hat{x} + c)^{\mathsf{T}} (-d) \geq 0.$$

Hence

$$(H\hat{x} + c)^{\mathsf{T}} d = 0 \qquad \text{for every } d \in \mathcal{N}(A).$$

Equivalently,

$$H\hat{x} + c \perp \mathcal{N}(A).$$

At an optimum, the gradient is orthogonal to every feasible direction.

From orthogonality to Lagrange multipliers

The fundamental subspace relation gives

$$\mathcal{N}(A)^{\perp} = \mathcal{R}(A^{\mathsf{T}}).$$

Thus

$$H\hat{x} + c \perp \mathcal{N}(A)$$

is equivalent to

$$H\hat{x} + c \in \mathcal{R}(A^{\mathsf{T}}).$$

Therefore, there exists a vector $\hat{u} \in \mathbb{R}^m$ such that

$$H\hat{x} + c = A^{\mathsf{T}}\hat{u}.$$

Together with primal feasibility,

$$A\hat{x} = b,$$

we obtain the optimality conditions

$$\boxed{H\hat{x} + c - A^{\mathsf{T}}\hat{u} = 0, \qquad A\hat{x} = b.}$$

The Lagrange system

The optimality conditions can be written as the linear system

$$\boxed{\begin{pmatrix} H & -A^{\mathsf{T}} \\ A & 0 \end{pmatrix} \begin{pmatrix} \hat{x} \\ \hat{u} \end{pmatrix} = \begin{pmatrix} -c \\ b \end{pmatrix}.}$$

The block matrix is symmetric after the equivalent sign convention

$$\begin{pmatrix} H & A^{\mathsf{T}} \\ A & 0 \end{pmatrix} \begin{pmatrix} \hat{x} \\ -\hat{u} \end{pmatrix} = \begin{pmatrix} -c \\ b \end{pmatrix}.$$

It is generally indefinite, even when the reduced Hessian is positive definite.

The Lagrange method avoids the explicit construction of a nullspace basis.

Why is u a Lagrange multiplier?

Define the Lagrangian

$$L(x, u) = \frac{1}{2} x^{\mathsf{T}} H x + c^{\mathsf{T}} x + c_0 - u^{\mathsf{T}} (Ax - b).$$

Stationarity with respect to x gives

$$Hx + c - A^{\mathsf{T}} u = 0.$$

The equality constraint is

$$Ax - b = 0.$$

Thus the Lagrange conditions are precisely

$$Hx + c - A^{\mathsf{T}} u = 0, \qquad Ax = b.$$

The multiplier supplies the normal component needed to balance the gradient.

A numerical example

Consider

$$\begin{aligned} &\text{minimize} && \tfrac{1}{2}x^{\mathsf{T}}Hx + c^{\mathsf{T}}x \\ &\text{subject to} && Ax = b, \end{aligned}$$

where

$$H = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 1 \\ 0 & 1 & 2 \end{pmatrix}, \qquad c = \begin{pmatrix} -3 \\ -3 \\ -8 \end{pmatrix}, \qquad A = \begin{pmatrix} 1 & 1 & 1 \end{pmatrix}, \qquad b = 2.$$

Choose the feasible point $\overline{x} = \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix}$ and the nullspace matrix $Z = \begin{pmatrix} -1 & -1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$.

The example: nullspace method

For the numerical example,

$$Z^{\mathsf{T}} H Z = \begin{pmatrix} 5 & 3 \\ 3 & 4 \end{pmatrix} \succ 0,$$

and

$$-Z^{\mathsf{T}}(H\overline{x} + c) = \begin{pmatrix} 4 \\ 9 \end{pmatrix}.$$

Thus

$$\begin{pmatrix} 5 & 3 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} \hat{v}_1 \\ \hat{v}_2 \end{pmatrix} = \begin{pmatrix} 4 \\ 9 \end{pmatrix},$$

which gives $\hat{v} = \begin{pmatrix} -1 \\ 3 \end{pmatrix}$. Therefore, $\hat{x} = \overline{x} + Z\hat{v} = \begin{pmatrix} 0 \\ -1 \\ 3 \end{pmatrix}$.

The example: Lagrange method

The Lagrange system is

$$\begin{pmatrix} 2 & 0 & 0 & -1 \\ 0 & 3 & 1 & -1 \\ 0 & 1 & 2 & -1 \\ 1 & 1 & 1 & 0 \end{pmatrix} \begin{pmatrix} \hat{x}_1 \\ \hat{x}_2 \\ \hat{x}_3 \\ \hat{u} \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \\ 8 \\ 2 \end{pmatrix}.$$

Solving gives

$$\hat{x} = \begin{pmatrix} 0 \\ -1 \\ 3 \end{pmatrix}, \qquad \hat{u} = -3.$$

The two methods give the same optimal solution.

The nullspace and Lagrange methods express the same optimality conditions in different coordinates.

Nullspace method or Lagrange method?

Nullspace method

- solves a system of size $(n-m) \times (n-m)$;
- makes feasible directions and the reduced Hessian explicit;
- requires a basis Z for $\mathcal{N}(A)$;
- may destroy sparsity because Z is often dense.

Lagrange method

- solves a system of size $(n+m) \times (n+m)$;
- avoids constructing Z;
- often preserves the sparsity of H and A;
- introduces the Lagrange multiplier u.

The best formulation depends on the dimensions and structure of the problem.

Main ideas

After this lecture, you should be able to:

- identify feasible directions with vectors in $\mathcal{N}(A)$;
- represent all feasible points as $x = \overline{x} + Zv$;
- construct a nullspace matrix from a partition $(A_B \; A_N)$;
- characterize convexity on the feasible set using $Z^{\mathsf{T}}HZ$;
- solve a small equality-constrained quadratic problem by the nullspace method;
- derive the Lagrange conditions from orthogonality to feasible directions;
- solve a small equality-constrained quadratic problem using the Lagrange system;
- compare the nullspace and Lagrange methods.

Equality constraints replace the full Hessian H by the reduced Hessian $Z^{\mathsf{T}}HZ$ along feasible directions.